MD KASHIF ALAM

Robotics Engineer | ROS2 • Computer Vision • Embedded Systems • AI Integration

CONTACT

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OBJECTIVE

Designing and developing autonomous robotic systems using AI, computer vision, and embedded control. Currently working with ROS2 for robot modeling, perception, and simulation, with a focus on building scalable robotic platforms for defense and commercial applications. Work samples are available on YouTube, LinkedIn, and GitHub.

EDUCATION

- **Bachelor of Science (BS),** Artificial Intelligence and CyberSecurity | IIT Patna
- **Computer Science (CSE)** | Institute of Innovation (PW), Bangalore

SKILLS

- **Programming Languages:** Python, C++, MicroPython
- **Computer Vision & AI:** OpenCV, YOLO, Mediapipe, TensorFlow
- **ROS2 & Simulation:** ROS2 (Humble), Gazebo, RViz, URDF Modeling
- **Embedded Systems:** Arduino, ESP32, STM32, Raspberry Pi, PCA9685
- **Sensor Interfacing:** MPU6050, Ultrasonic, LIDAR, Camera Modules
- **Hardware Communication:** UART (Serial Communication), I²C, SPI, CAN (basic)
- **Hardware Control:** Servo & Motor Control, PID Tuning, PWM
- **Development Tools:** VS Code, Arduino IDE, GitHub, Linux, WSL, Processing

TECHNICAL EXPERIENCE

- **AI-Powered Gesture-Controlled Robotic Hand**
    - Developed a robotic hand controlled via **OpenCV** and **Mediapipe.**
    - Integrated **servo motors** with **real-time vision processing** for accurate hand movements.
- **Sign-Controlled Light**
    - Implemented **gesture recognition** to control lighting — *Thumbs Up → Light ON, Thumbs Down → Light OFF.*
- **AI-Powered Mentor**
    - Built an AI mentor application with a **floating interface** to capture screen or voice input, process via **AI APIs**, and respond intelligently in **voice or text**.
- **Robotic Arm**
    - Developed a **robotic arm** capable of **precise 3D point movement** using multiple **servo motors** for accuracy and smooth control.
- **Surveillance Drone**
    - Designed and built a drone equipped with **real-time video streaming** for **aerial monitoring and surveillance**.
- **Quadruped Robotic Dog (Ongoing)**
    - Engineering an **autonomous quadruped robot** capable of walking, running, and turning, with future integration of **AI-based navigation, mapping, and localization.**
- **Ornithopter (Ongoing)**
    - Engineered a **gearbox-driven flapping-wing mechanism** replicating **avian flight**, and progressing the **control system** for **stable**, **semi-autonomous aerial surveillance.**